



# What Factors Impact a Movie's Revenue?

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STAT 183 Group 1  
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- 
- The original dataset *movies* was downloaded from Kaggle.
  - This project aims to deepen the understanding of how different factors may impact a movie's revenue.
  - Hypotheses:
    1. Different attributes of a feature movie significantly impacts its revenue.
    2. A movie's budget and genre contribute more significantly than others.
  - Other questions of interest:
    1. Which genres generated the highest revenue?
    2. Which movies had the highest revenue or profit?
    3. Does budget affect the runtime of a movie?

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- 
- Original data set
    - 722,463 samples & 20 variables
  - Data cleaning criteria
    - Omit unnecessary variables
    - Omit all NA's and 0's
    - *budget & revenue*  $\geq$  \$10,000
    - *runtime*  $\geq$  40 minutes
    - *vote count*  $\geq$  95 votes
    - *vote average*  $\geq$  0
    - **The cleaned dataset now has 6,804 samples & 8 variables**

- 
- The dataset has too many genres and original languages
    - Top 2 *genres*: Drama (1,501) & Comedy (1,401)
    - Top 2 *original languages*: English (5,945) & French (217)
    - Divided *revenue* (measured in dollars) by 1,000
  - **The dataset now has 2,624 samples & 8 variables → will be used for model building**

|    | genres | original_language | popularity | budget | revenue     | runtime | vote_average | vote_count |
|----|--------|-------------------|------------|--------|-------------|---------|--------------|------------|
| 1  | Drama  | en                | 3994.342   | 75000  | 269000.000  | 116     | 7.262        | 1129       |
| 2  | Drama  | en                | 1329.764   | 100000 | 322300.000  | 157     | 7.300        | 1211       |
| 3  | Comedy | en                | 604.239    | 125000 | 632221.453  | 117     | 7.212        | 2684       |
| 4  | Comedy | en                | 441.504    | 145000 | 1445638.421 | 114     | 7.140        | 7064       |
| 5  | Comedy | en                | 430.749    | 40000  | 13280.000   | 140     | 7.095        | 3584       |
| 6  | Drama  | en                | 423.702    | 100000 | 952000.000  | 181     | 8.111        | 7611       |
| 7  | Comedy | en                | 395.745    | 45000  | 82995.526   | 104     | 6.889        | 275        |
| 8  | Drama  | en                | 329.063    | 50000  | 214215.889  | 130     | 6.990        | 4667       |
| 9  | Drama  | en                | 320.743    | 37000  | 173567.581  | 133     | 7.405        | 6549       |
| 10 | Drama  | en                | 310.154    | 1000   | 42000.000   | 111     | 7.684        | 122        |
| 11 | Drama  | en                | 269.007    | 40000  | 175302.354  | 139     | 8.189        | 11951      |
| 12 | Drama  | en                | 195.639    | 3000   | 4109.095    | 117     | 5.600        | 135        |
| 13 | Drama  | en                | 188.422    | 55000  | 368307.760  | 105     | 6.699        | 7194       |
| 14 | Drama  | en                | 168.905    | 40000  | 571006.128  | 125     | 5.900        | 10758      |

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- **Response variable**

- **revenue** (measured in thousands of dollars): the amount of money a movie raised by ticket sales, numerical

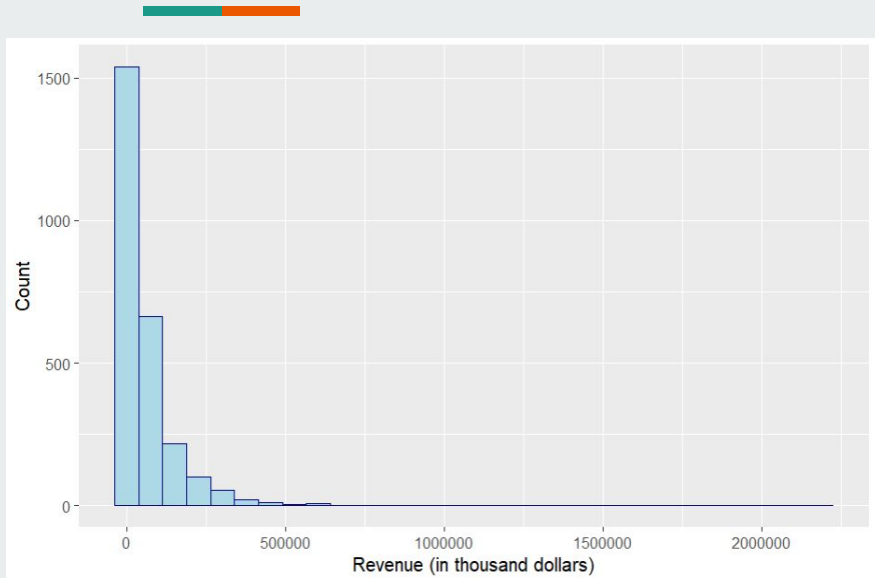
- **Predictor variables**

- **genres** (Drama or Comedy): genre of the movie, categorical
- **original language** (en or fr): the language in which the movie was first released, categorical
- **popularity**: the amount of times a movie's page has been visited according to TMDB metric, numerical
- **budget** (measured in thousand dollars): the total projected costs of making a movie, numerical
- **runtime** (measured in minutes): duration of the movie, numerical
- **vote average** (on a scale from 0-10): average of votes given by TMDB users, numerical
- **vote count**: total number of TMDB votes a movie received, numerical

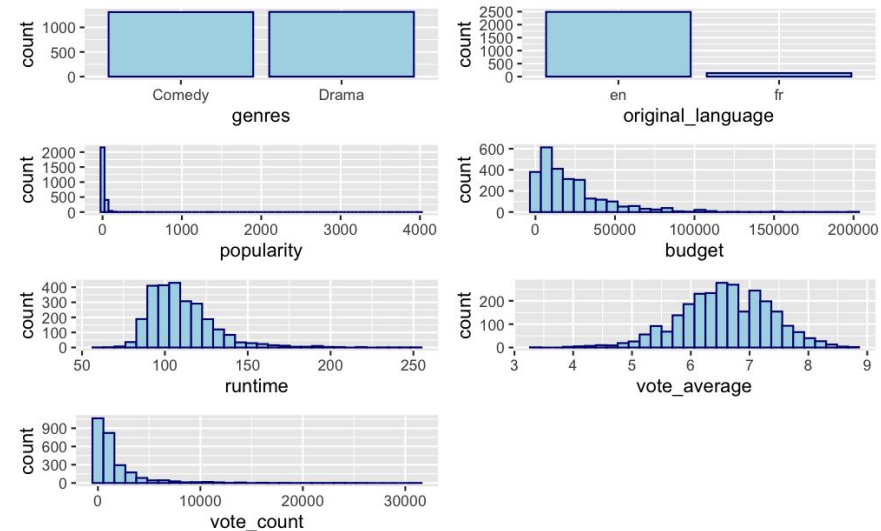
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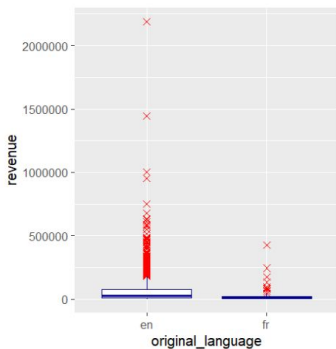
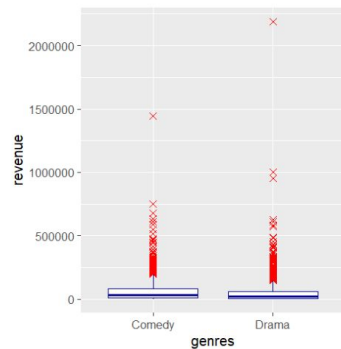
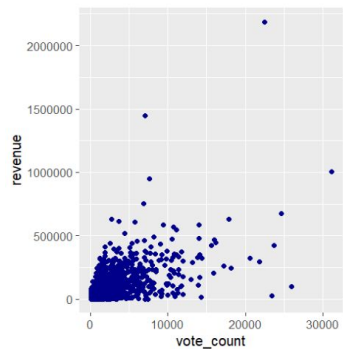
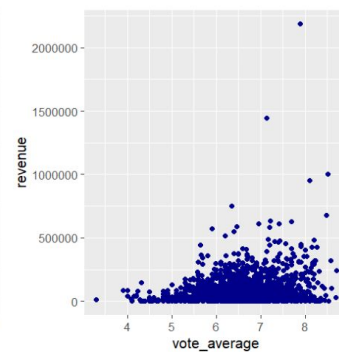
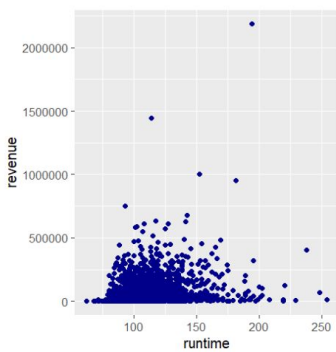
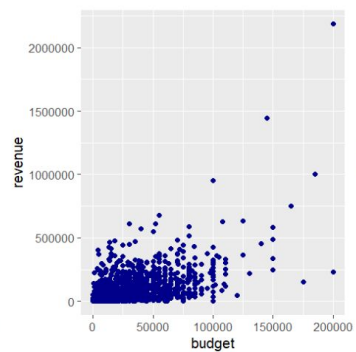
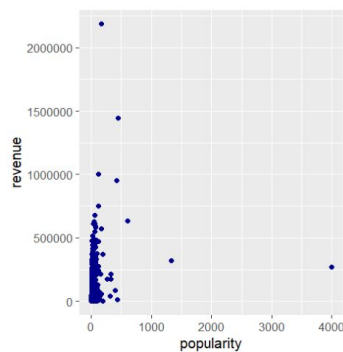
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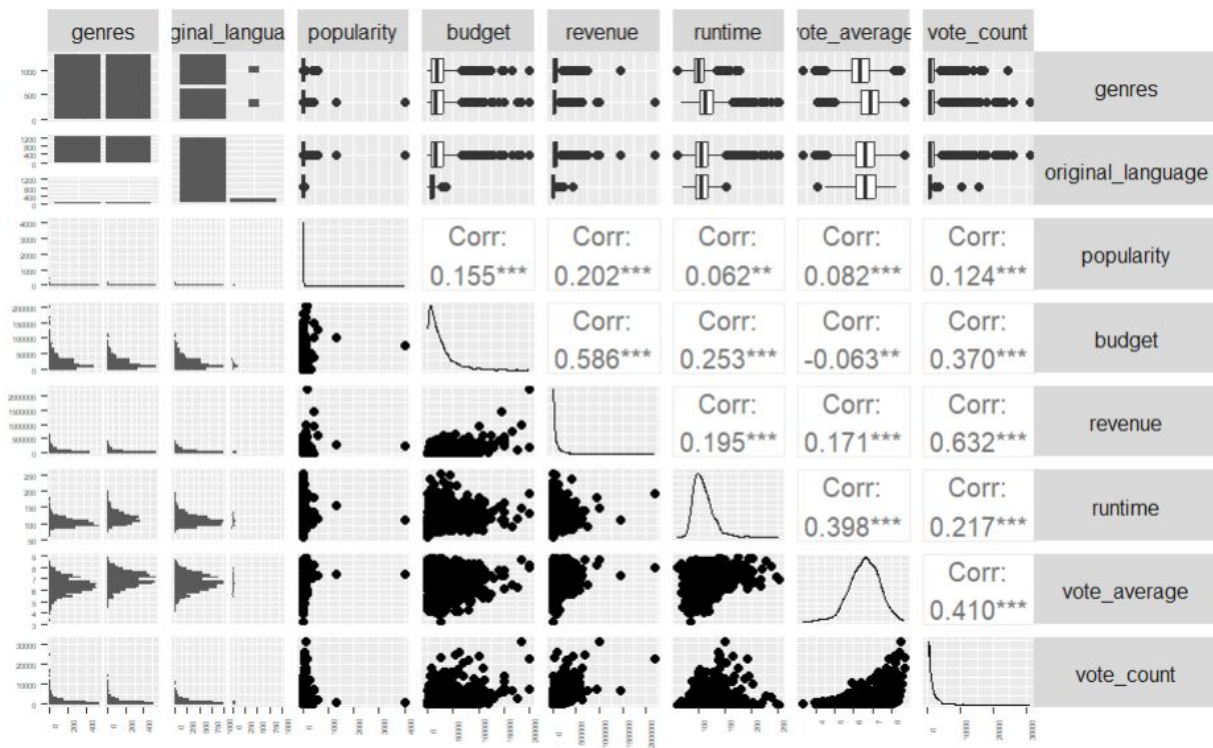


Histogram of responsive variable, *revenue*  
(measured in thousand dollars)



Bar plots for categorical predictors and  
histograms for numerical predictors





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1. Stepwise regression

- Final predictors: *vote count*, *budget*, *popularity*, *genres*
- Adjusted  $R^2$ : 0.5549

2. Check model assumptions

- Passes linearity check
- Does not pass independence check

3. Convert *genres* and *original language* into binary variables

4. Check interaction between *genre* and *popularity*

- Present but does not seem to significantly impact the model

5. Check the effect of budget

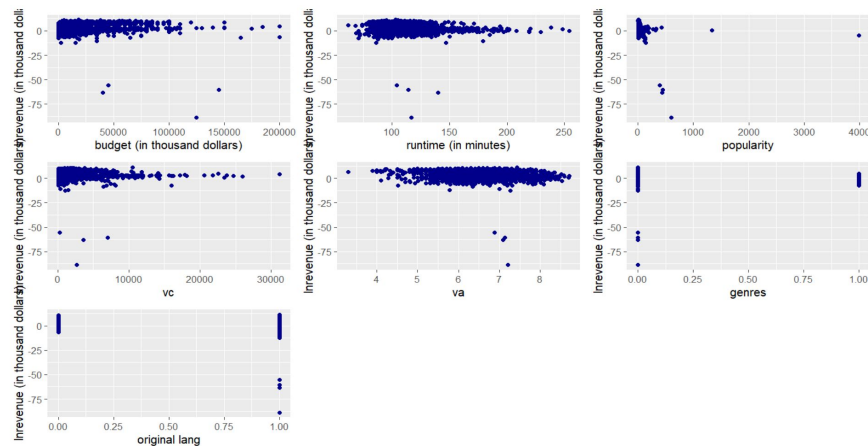
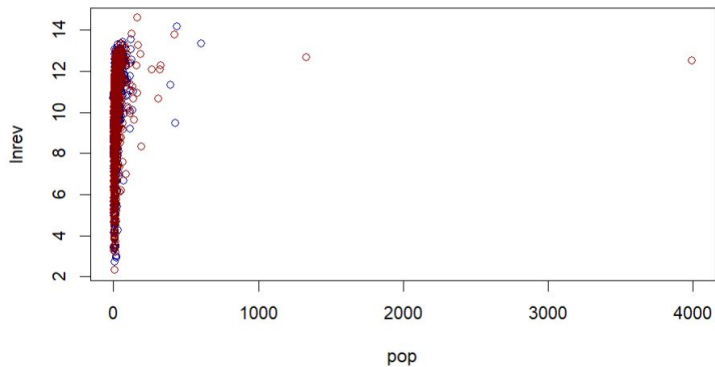
- Present but does not seem to significantly impact the model

**\* Decides to transform *revenue* (y) using log transformation**

\* *revenue* → *lnrev*

Check interaction between *genres* and *popularity*

```
lm1 <- lm(lnrev ~ 0 + pop * genre)
y1 <- lm1$residuals
```

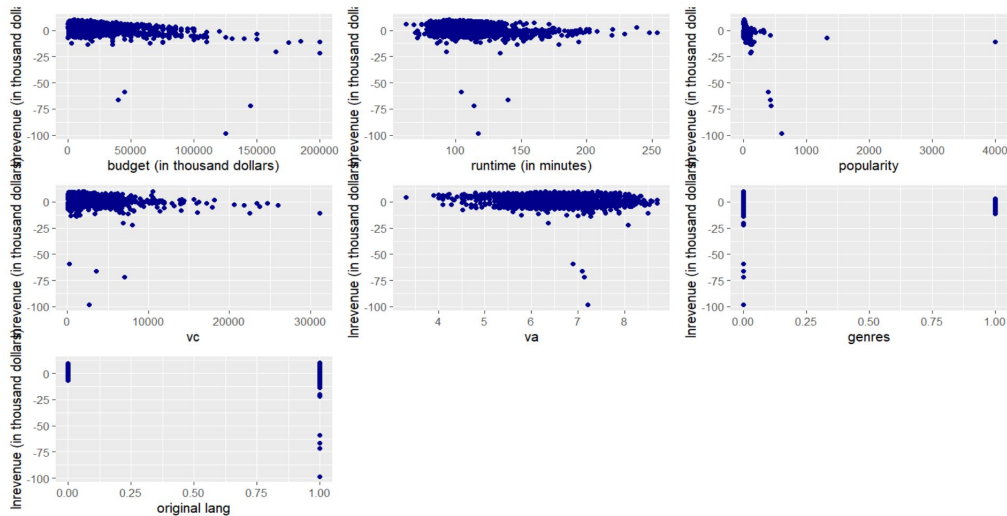




\* *revenue* → *lnrev*

Check the effect of *budget*

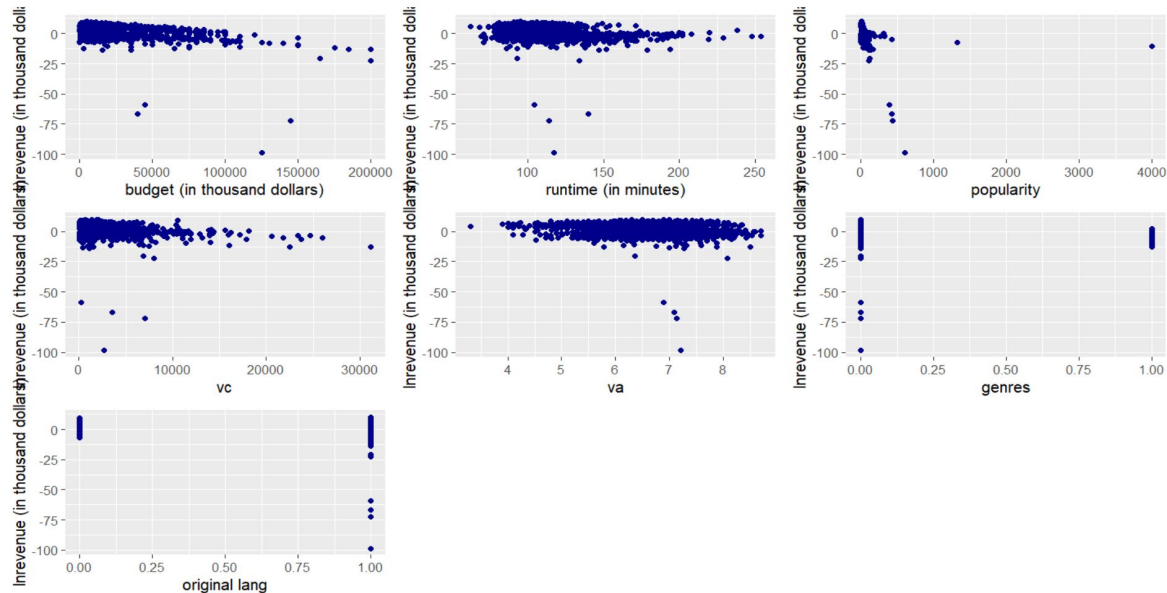
```
lm2 <- lm(y1 ~ 0 + budget)
y2 <- lm2$residuals
```



\* *revenue* → *lnrev*

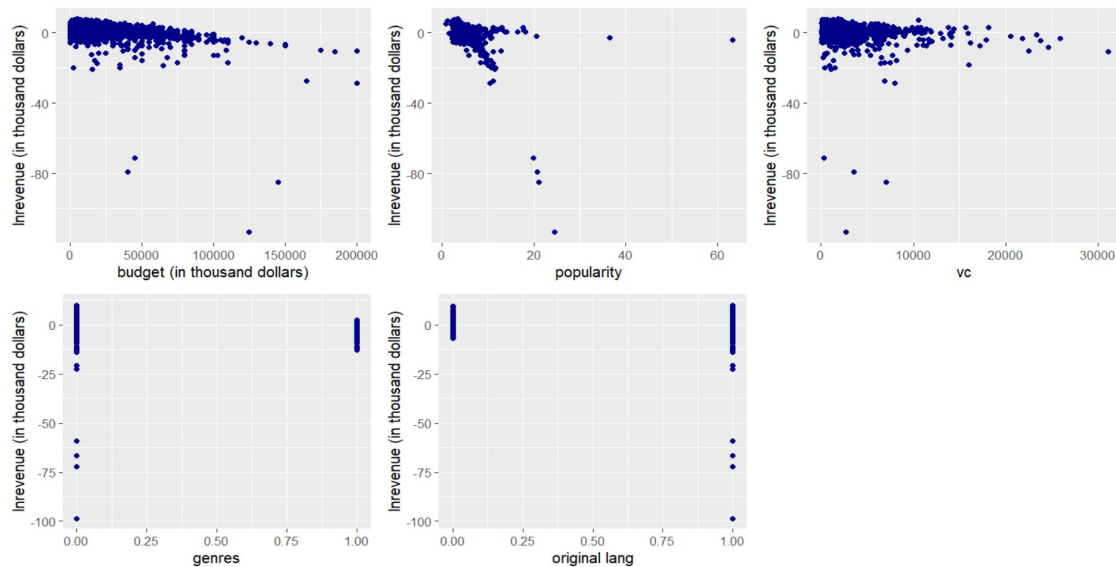
Check the effect of *vote count* (*vc*)

```
lm3 <- lm(y2 ~ 0 + vc)
y3 <- lm3$residuals
```



\* *revenue* → *lnrev*

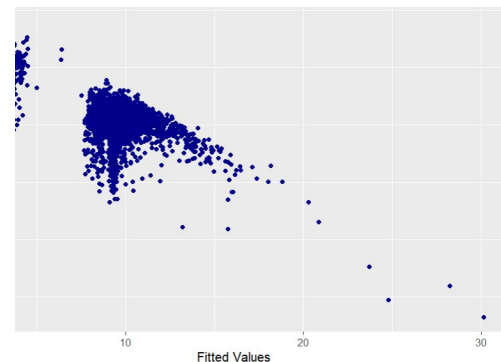
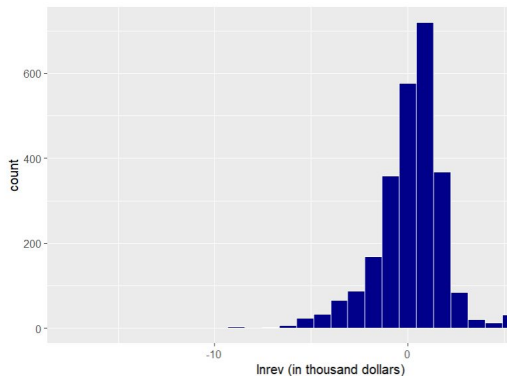
\* *pop* → *popsqrt*



# Model 1

```
lm5 <- lm(lnrev ~ 0 + budget + popsqrt * genre + o_lang + vc)
summary(lm5)
```

```
## Call:
## lm(formula = lnrev ~ 0 + budget + popsqrt * genre + o_lang +
##     vc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.7908  -0.6587   0.4173   1.2191   8.9885
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## budget          2.637e-05  1.936e-06  13.620 < 2e-16 ***
## popsqrt          8.531e-01  3.021e-02  28.237 < 2e-16 ***
## genre           3.320e+00  1.622e-01  20.468 < 2e-16 ***
## o_lang          5.516e+00  1.273e-01  43.335 < 2e-16 ***
## vc              1.363e-04  1.825e-05   7.468 1.1e-13 ***
## popsqrt:genre  -7.775e-01  3.581e-02 -21.715 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.07 on 2618 degrees of freedom
## Multiple R-squared:  0.9579, Adjusted R-squared:  0.9578
## F-statistic: 9929 on 6 and 2618 DF, p-value: < 2.2e-16
```

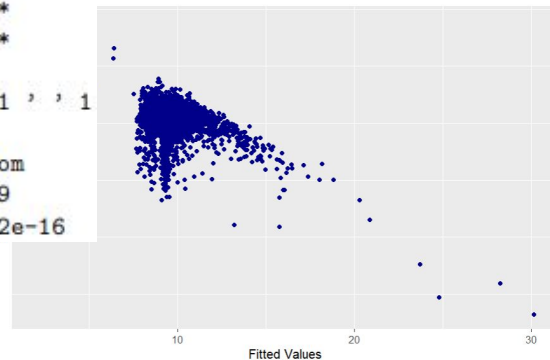
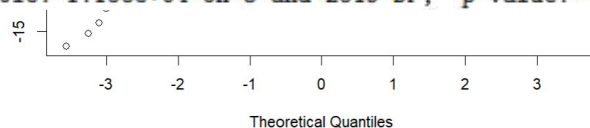
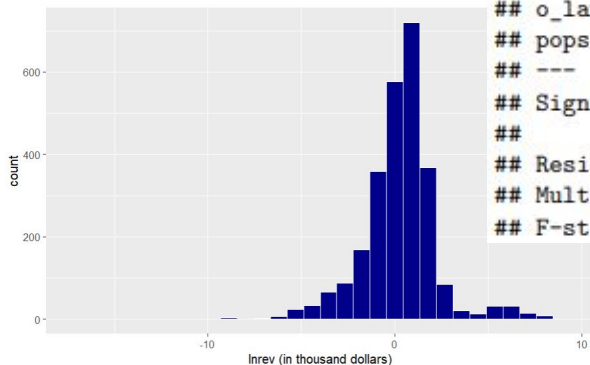


# Model 2

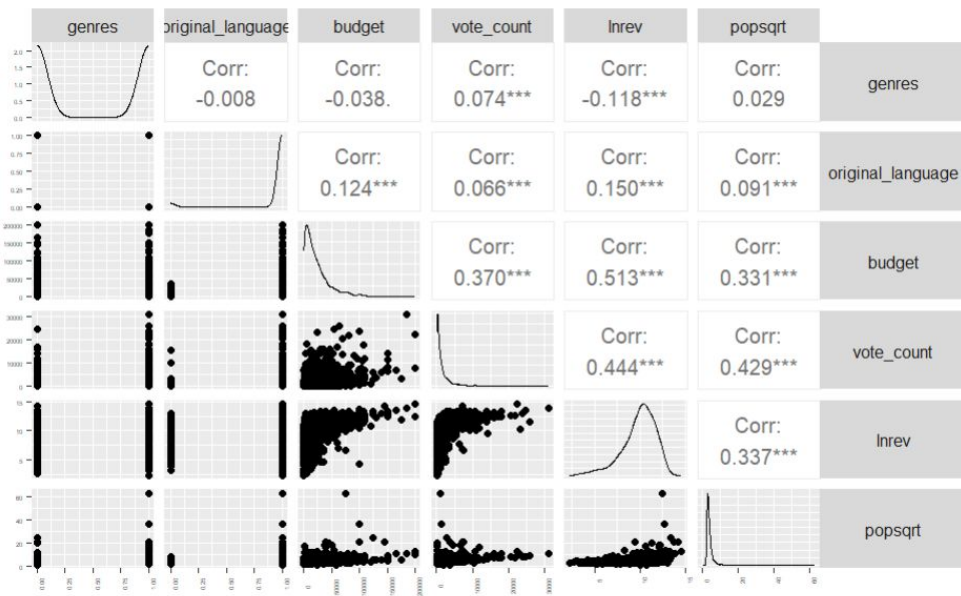
```
lm6 <-  
summary
```

```
## Call:  
## lm(formula = lnrev ~ 0 + budget + popsqrt * genre + o_lang)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -18.0169  -0.6501   0.3745   1.2554   9.0827   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## budget          3.027e-05  1.883e-06  16.07  <2e-16 ***  
## popsqrt          9.017e-01  2.981e-02  30.25  <2e-16 ***  
## genre            3.346e+00  1.638e-01  20.42  <2e-16 ***  
## o_lang            5.425e+00  1.280e-01  42.38  <2e-16 ***  
## popsqrt:genre    -7.738e-01  3.617e-02 -21.39  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 2.092 on 2619 degrees of freedom  
## Multiple R-squared:  0.957, Adjusted R-squared:  0.9569   
## F-statistic: 1.166e+04 on 5 and 2619 DF,  p-value: < 2.2e-16
```

```
e + o_lang)
```



# Model 1 diagnostics



|               | budget   | popsqrt   | genre    | o_lang   | vc       |
|---------------|----------|-----------|----------|----------|----------|
| budget        | 2.401224 | 12.518903 | 8.065951 | 9.415968 | 1.863471 |
| popsqrt:genre | 9.708687 |           |          |          |          |

Step: AIC=3824.58

lnrev ~ o\_lang + popsqrt + budget + vc + genre + popsqrt:genre - 1

|                 | Df | Sum of Sq | RSS   | AIC    | F value  | Pr(>F)        |
|-----------------|----|-----------|-------|--------|----------|---------------|
| <none>          |    |           | 11220 | 3824.6 |          |               |
| - vc            | 1  | 239.0     | 11459 | 3877.9 | 55.773   | 1.102e-13 *** |
| - budget        | 1  | 795.0     | 12015 | 4002.2 | 185.496  | < 2.2e-16 *** |
| - popsqrt:genre | 1  | 2020.9    | 13240 | 4257.2 | 471.556  | < 2.2e-16 *** |
| - o_lang        | 1  | 8047.9    | 19268 | 5241.5 | 1877.913 | < 2.2e-16 *** |
| ---             |    |           |       |        |          |               |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

# Final model comparison & selection

---

```
# model option 1: all predictor variables in linear form
op1 <- lm(lnrev ~ 0 + genre + budget + vc + popsqrt + o_lang)

# model option 2: no categorical vars
op2 <- lm(lnrev ~ 0 + budget + popsqrt * genre)

# model option 3: lm5 aka the stepwise final model
op3 <- lm(lnrev ~ 0 + budget + popsqrt * genre + vc + o_lang)

# compare op1 and op2 using F-statistic
anova(op1, op2)

# compare op2 and op3 using F-statistic
anova(op2, op3)
```

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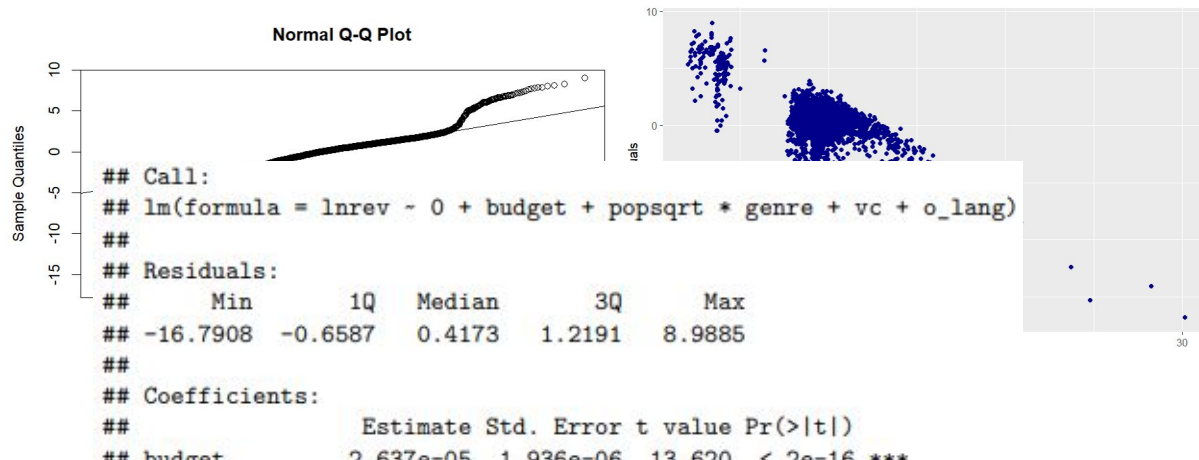
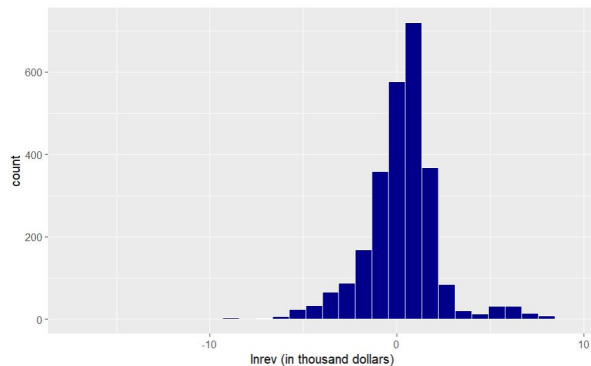
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# Final model

```
final.model <- lm(lnrev ~ 0 + budget + popsqrt * genre + vc + o_lang)
summary(final.model)
```

|               | budget   | popsqrt   | genre    | vc       | o_lang   |
|---------------|----------|-----------|----------|----------|----------|
|               | 2.401224 | 12.518903 | 8.065951 | 1.863471 | 9.415968 |
| popsqrt:genre |          |           |          |          |          |
|               |          |           |          |          | 9.708687 |





Final model:  $\hat{Y} = 2.6366e^{-5}x_1 + 8.5311e^{-1}x_2 + 3.3198x_3 - 7.7753e^{-1}x_4 + 1.3633e^{-4}x_5 + 5.5155x_6$

Where:

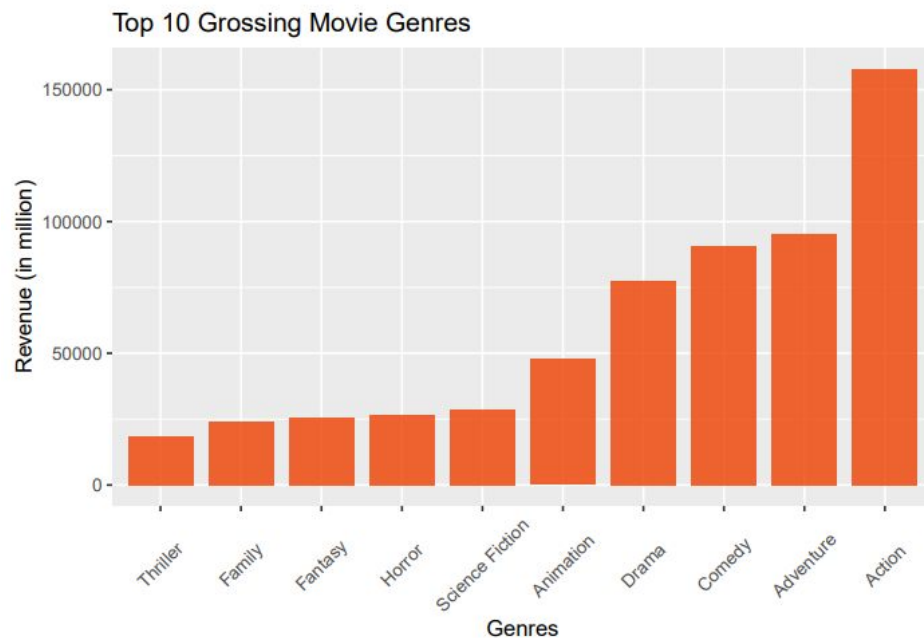
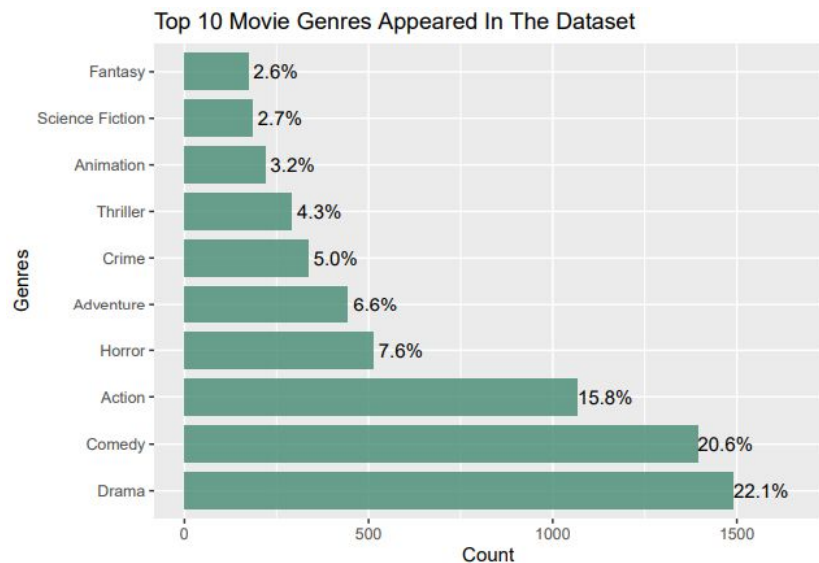
- $\hat{Y} = \ln \text{revenue}$
- $x_1 = \text{budget}$
- $x_2 = \text{popsqrt}$
- $x_3 = \text{genre}$
- $x_4 = \text{popsqrt} * \text{genre}$
- $x_5 = \text{vote count}$
- $x_6 = \text{original language}$

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# Which Genre Generated the Highest Revenue?

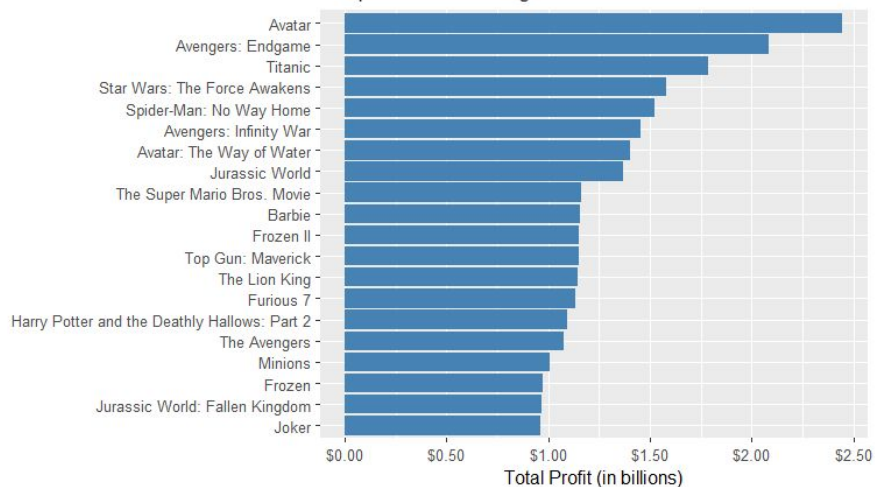


# Which Movie Generated the Highest Revenue or Profit?

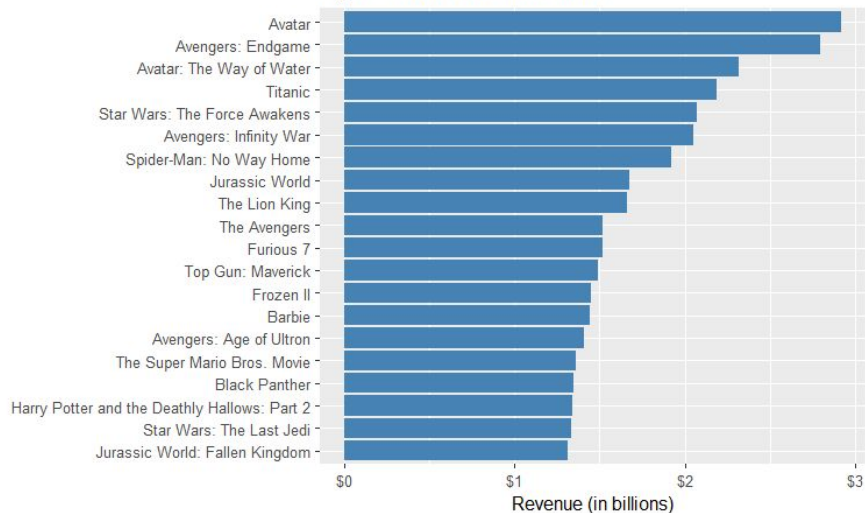


## Top Profit Movies

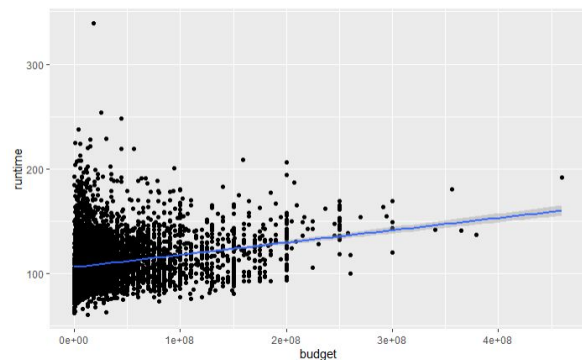
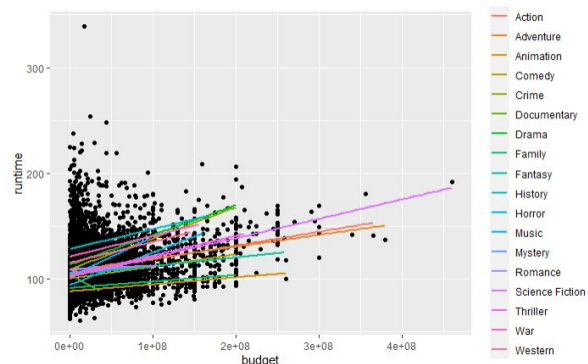
total profit = revenue-budget



## Highest Revenue Generated



# Does Budget Affect the Runtime of Movies?



| genres          | Intercept | Slope         |
|-----------------|-----------|---------------|
| Action          | 105.50643 | 1.307857e-07  |
| Adventure       | 108.01356 | 1.126079e-07  |
| Animation       | 88.33316  | 6.579880e-08  |
| Comedy          | 100.30342 | 1.143451e-07  |
| Crime           | 109.22148 | 2.938780e-07  |
| Documentary     | 104.47996 | -4.381655e-07 |
| Drama           | 113.58229 | 2.811547e-07  |
| Family          | 91.48680  | 6.068321e-08  |
| Fantasy         | 104.00737 | 8.127584e-08  |
| History         | 128.25633 | 1.886335e-07  |
| Horror          | 94.31043  | 3.070825e-07  |
| Music           | 102.31549 | 3.722074e-07  |
| Mystery         | 105.95024 | 7.115128e-08  |
| Romance         | 109.20149 | 3.376121e-08  |
| Science Fiction | 102.08929 | 1.832165e-07  |
| TV Movie        | 89.00000  | NA            |
| Thriller        | 103.22143 | 1.888751e-07  |
| War             | 121.12549 | 1.954602e-07  |
| Western         | 115.10759 | 1.995560e-07  |

Residuals:

| Min     | 1Q      | Median | 3Q    | Max     |
|---------|---------|--------|-------|---------|
| -46.728 | -13.492 | -3.493 | 9.743 | 230.684 |

Coefficients:

|             | Estimate  | Std. Error | t value | Pr(> t )   |
|-------------|-----------|------------|---------|------------|
| (Intercept) | 1.062e+02 | 3.079e-01  | 344.87  | <2e-16 *** |
| budget      | 1.176e-07 | 5.766e-09  | 20.39   | <2e-16 *** |

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# Conclusion



We have obtained a final model that statistically significant with extremely high adjusted  $R^2$  value.

The final model suggests that a movie's revenue is affected by its budget, popularity, genre, vote count, original language and the interaction between the popularity and genre.

Although we have a high R-squared value, it did not pass the Shapiro-Wilk test meaning there are some things we could've done better for future work. We could either use the full data, including all movie genres and original languages. Also, an alternative solution could be for us to gather our own data of movie votes since the origin of the dataset votes are unknown.



# Limitations



- Not being able to adjust the data enough to satisfy the Shapiro-Wilk test
  - Cannot be fully certain the data is approximately normal though it is sufficient enough proven by qqnorm and histogram of residuals
- No sources for where the voting came from
  - Potential bias with the voters
- Not enough time to deep dive into the analysis of movies involving genres other than Comedy and Drama nor languages other than English and French
- *popsqrt* in our final model is close but does not fulfill the VIF condition as it is 11.93 → has a very small multicollinearity issue